Math 15 — Unit 2: Assignments

The assignment from every lesson in the booklet. Printed work is not tracked on the website.

Lesson 1 Assignment *Show your work.*

Part A — Vocabulary and Evaluating

1. State the base and the exponent in each power.
(a) 6^3 (b) y^8 (c) 4^n (d) $(-p)^5$ (e) 9^2

2. Write each power as a repeated multiplication.
(a) 5^3 (b) w^4 (c) $(-2)^5$ (d) q^6

3. Evaluate each power.
(a) 3^5 (b) 6^2 (c) 2^6 (d) 5^3 (e) 10^3 (f) 4^4

4. Evaluate without a calculator. Pay close attention to the brackets.
(a) -7^2 (b) $(-7)^2$ (c) -4^3 (d) $(-4)^3$ (e) -2^4 (f) $(-2)^4$

5. In your own words, explain why -5^2 and $(-5)^2$ are not equal.

6. Evaluate each algebraic power for the given value of the variable.
(a) a^2, $a = 4$ (b) b^3, $b = 3$ (c) c^2, $c = -4$ (d) d^4, $d = -1$ (e) e^3, $e = -3$

Part B — The Product Law

7. Expand $4^2 \times 4^3$ into repeated multiplication, count the total factors of 4, rewrite as a single power, then evaluate it.

8. Use the Product Law to simplify, then evaluate.
(a) $3^2 \times 3^3$ (b) $6^2 \times 6$ (c) $2^3 \times 2^1 \times 2^2$ (d) $5^2 \times 5^2$ (e) $10^3 \times 10^2$

9. Use the Product Law to simplify.
(a) $x^3 \times x^4$ (b) $y^6 \times y$ (c) $m^2 \times m^3 \times m^4$ (d) $k^5 \times k^5$ (e) $p \times p^7$

10. Use the Product Law to simplify each expression involving coefficients.
(a) $3x^4 \times 5x^2$ (b) $5a^2 \times a^4$ (c) $3m \times 6m^5$ (d) $8b^3 \times 2b^2$ (e) $6n^3 \times n^2$

11. Find the value of x that makes each statement true.
(a) $a^3 \times a^x = a^9$ (b) $b^x \times b^4 = b^{10}$ (c) $c^5 \times c^x = c^{12}$

12. *(Multiple Choice)* Which of the following is equivalent to $m^4 \times m^3$? A. m^7 B. m^{12} C. m^1 D. $2m^7$

13. A rectangular prism has a square base with side length x^2 and a height of x^3. Write an expression for its volume as a single power of x.

Answer Key. 1. (a) base 6, exp 3 (b) base y, exp 8 (c) base 4, exp n (d) base $-p$, exp 5 (e) base 9, exp 2. **2.** (a) $5\times5\times5$ (b) $w\times w\times w\times w$ (c) five factors of (-2) (d) six factors of q. **3.** (a) 243 (b) 36 (c) 64 (d) 125 (e) 1000 (f) 256. **4.** (a) -49 (b) 49 (c) -64 (d) -64 (e) -16 (f) 16. **5.** No brackets: $-5^2 = -25$; brackets make -5 the base: $(-5)^2 = 25$. **6.** (a) 16 (b) 27 (c) 16 (d) 1 (e) -27. **7.** $4^5 = 1024$. **8.** (a) $3^5 = 243$ (b) $6^3 = 216$ (c) $2^6 = 64$ (d) $5^4 = 625$ (e) $10^5 = 100\,000$. **9.** (a) x^7 (b) y^7 (c) m^9 (d) k^{10} (e) p^8. **10.** (a) $15x^6$ (b) $5a^6$ (c) $18m^6$ (d) $16b^5$ (e) $6n^5$. **11.** (a) 6 (b) 6 (c) 7. **12.** A. **13.** x^7.

Lesson 2 Assignment *Show your work.*

Part A — The Quotient Law

1. Fill in each blank, then evaluate. (The bases are the same, so subtract the exponents.)
(a) $6^7 \div 6^4 = 6^{__} = _____$ (b) $9^5 \div 9^3 = 9^{__} = _____$

2. Evaluate each expression using the Quotient Law.
(a) $5^6 \div 5^3$ (b) $\dfrac{8^5}{8^2}$ (c) $\dfrac{(-4)^7}{(-4)^5}$ (d) $2^9 \div 2^4$

3. Simplify each expression using the Quotient Law.
(a) $x^8 \div x^3$ (b) $\dfrac{m^{11}}{m^6}$ (c) $n^7 \div n^2$ (d) $\dfrac{w^{10}}{w^4}$

4. Simplify. Divide the coefficients, then use the Quotient Law on the variable part.
(a) $15x^7 \div 3x^2$ (b) $\dfrac{24y^9}{6y^4}$ (c) $21a^8 \div 7a^3$ (d) $\dfrac{16b^6}{4b^2}$

Part B — Power of a Power

5. Fill in each blank, then evaluate. (A power is raised to another power, so multiply the exponents.)
(a) $\left(4^2\right)^3 = 4^{__} = _____$ (b) $\left(2^3\right)^3 = 2^{__} = _____$

6. Evaluate each expression using the Power of a Power Law.
(a) $\left(3^2\right)^2$ (b) $\left(2^2\right)^4$ (c) $\left(5^2\right)^2$ (d) $\left(7^2\right)^2$

7. Simplify each expression using the Power of a Power Law.
(a) $\left(x^3\right)^5$ (b) $\left(m^6\right)^2$ (c) $\left(k^4\right)^4$ (d) $\left(t^2\right)^7$

Part C — Mixing the Laws

8. Decide which exponent law applies, then simplify.
(a) $y^4 \times y^3$ (b) $\dfrac{z^9}{z^4}$ (c) $\left(c^3\right)^4$ (d) $h^6 \times h^2 \div h^3$

9. Use both the Quotient Law and the Power of a Power Law to simplify.
(a) $\dfrac{\left(y^2\right)^4}{y^3}$ (b) $\left(3^3\right)^2 \div 3^4$ (c) $\dfrac{\left(a^4\right)^3}{\left(a^2\right)^2}$

10. Use the Quotient Law to state the value of x in each equation.
(a) $c^x \div c^4 = c^9$ (b) $\dfrac{d^{15}}{d^x} = d^6$ (c) $g^x \div g^7 = g^{12}$

11. Use the Power of a Power Law to state the value of x in each equation.
(a) $(a^x)^3 = a^{15}$ (b) $\left(b^5\right)^x = b^{30}$

12. Explain, using factors, why $\left(k^2\right)^3 \neq k^2 \times k^3$.

13. *(Multiple Choice)* Which of the following is equivalent to $\left(x^3\right)^5$? A. x^8 B. x^{15} C. x^{125} D. x^{243}

14. *(Numerical Response)* If $\dfrac{p^{18}}{p^n} = p^6$, where n is a whole number, then the value of n is _____.

Answer Key. 1. (a) $6^3 = 216$ (b) $9^2 = 81$. **2.** (a) $5^3 = 125$ (b) $8^3 = 512$ (c) $(-4)^2 = 16$ (d) $2^5 = 32$. **3.** (a) x^5 (b) m^5 (c) n^5 (d) w^6. **4.** (a) $5x^5$ (b) $4y^5$ (c) $3a^5$ (d) $4b^4$. **5.** (a) $4^6 = 4096$ (b) $2^9 = 512$. **6.** (a) $3^4 = 81$ (b) $2^8 = 256$ (c) $5^4 = 625$ (d) $7^4 = 2401$. **7.** (a) x^{15} (b) m^{12} (c) k^{16} (d) t^{14}. **8.** (a) y^7 (b) z^5 (c) c^{12} (d) h^5. **9.** (a) y^5 (b) $3^2 = 9$ (c) a^8. **10.** (a) $x = 13$ (b) $x = 9$ (c) $x = 19$. **11.** (a) $x = 5$ (b) $x = 6$. **12.** $\left(k^2\right)^3 = k^2 \times k^2 \times k^2 = k^6$, but $k^2 \times k^3 = k^5$; $k^6 \neq k^5$. **13.** B. **14.** 12.

Lesson 3 Assignment *Show your work.*

Part A — The Two New Laws

1. Complete each exponent law statement.

(a) $(ab)^n =$ _____ (b) $\left(\frac{a}{b}\right)^n =$ _____ $(b \neq 0)$

2. Simplify using the Product Law, the Quotient Law, or the Power of a Power Law.

(a) $x^4 \times x^5$ (b) $y^9 \div y^3$ (c) $(m^3)^2$ (d) $n^6 \times n \times n^2$

3. State whether the Power of a Product Law or the Power of a Quotient Law applies, then simplify.

(a) $(2y)^3$ (b) $\left(\frac{m}{4}\right)^2$ (c) $(5ab)^2$ (d) $\left(\frac{a}{b}\right)^6$

4. Use the Power of a Product Law to simplify.

(a) $(2x)^3$ (b) $(5y)^2$ (c) $(-3m)^2$ (d) $(4ab)^2$ (e) $(10w)^3$ (f) $(-2xy)^3$

5. Use the Power of a Quotient Law to simplify.

(a) $\left(\frac{2}{3}\right)^2$ (b) $\left(\frac{x}{4}\right)^2$ (c) $\left(\frac{7}{y}\right)^2$ (d) $\left(\frac{m}{n}\right)^5$ (e) $\left(\frac{a}{2}\right)^3$ (f) $\left(\frac{3}{b}\right)^4$

6. Evaluate completely (numeric bases only), and use the Power of a Product or Power of a Quotient Law to check your work.

(a) $(2 \times 3)^3$ (b) $\left(\frac{10}{2}\right)^2$ (c) $(-2 \times 5)^2$ (d) $\left(\frac{12}{4}\right)^2$

Part B — Combining Laws

7. Simplify each expression using two exponent laws.

(a) $(3x)^2 \times x^4$ (b) $(y^2)^3 \div y^3$ (c) $(3m)^3 \times m^2$ (d) $(a^2)^4 \div a^5$ (e) $(4n^2)^2$ (f) $\left(\frac{p}{2}\right)^2 \times p^3$

8. These combine three or more exponent laws. Simplify each expression completely.

(a) $(3x)^2(x^2)^3$ (b) $\frac{(2m^3)^2}{m^4}$ (c) $\left(\frac{3a^2}{b}\right)^2$ (d) $\frac{(2x^2)^3}{x^4}$

9. State the value of the unknown.

(a) $(2x)^n = 8x^3$, find n. (b) $(ax)^2 = 9x^2$, find the positive value of a. (c) $\left(\frac{x}{k}\right)^2 = \frac{x^2}{25}$, find the positive value of k.

10. (a) In your own words, explain the difference between $3x^2$ and $(3x)^2$. Then evaluate both expressions when $x = 2$.

(b) Explain why the Power of a Quotient Law, $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$, requires $b \neq 0$.

11. *(Multiple Choice)* Which of the following is equivalent to $(4y)^2$? A. $8y^2$ B. $16y^2$ C. $4y^2$ D. $16y$

12. *(Numerical Response)* If $(3m^2)^2 = km^4$, the value of k is _____.

Answer Key. 1. (a) $a^n b^n$ (b) $\frac{a^n}{b^n}$. **2.** (a) x^9 (b) y^6 (c) m^6 (d) n^9. **3.** (a) Product; $8y^3$ (b) Quotient; $\frac{m^2}{16}$ (c) Product; $25a^2b^2$ (d) Quotient; $\frac{a^6}{b^6}$. **4.** (a) $8x^3$ (b) $25y^2$ (c) $9m^2$ (d) $16a^2b^2$ (e) $1000w^3$ (f) $-8x^3y^3$. **5.** (a) $\frac{4}{9}$ (b) $\frac{x^2}{16}$ (c) $\frac{49}{y^2}$ (d) $\frac{m^5}{n^5}$ (e) $\frac{a^3}{8}$ (f) $\frac{81}{b^4}$. **6.** (a) 216 (b) 25 (c) 100 (d) 9. **7.** (a) $9x^6$ (b) y^3 (c) $27m^5$ (d) a^3 (e) $16n^4$ (f) $\frac{p^5}{4}$. **8.** (a) $9x^8$ (b) $4m^2$ (c) $\frac{9a^4}{b^2}$ (d) $8x^2$. **9.** (a) $n = 3$ (b) $a = 3$ (c) $k = 5$. **10.** (a) In $3x^2$ only x is squared; in $(3x)^2$ the whole quantity is. At $x = 2$: 12 and 36. (b) A zero denominator makes the fraction undefined. **11.** B. **12.** 9.

Lesson 4 Assignment *Show your work.*

Part A — Warm-Up and Two-Law Combinations

1. Simplify each expression using just *one* law (a quick warm-up review).
(a) $x^3 \times x^4$ (b) $y^8 \div y^3$ (c) $(a^2)^4$ (d) $(2b)^3$

2. Simplify. Apply the power law first, then the product law.
(a) $(2x^2)^2 \times x^3$ (b) $(3a^3)^2 \times 2a^2$

3. Simplify. Apply the power law first, then the quotient law.
(a) $\dfrac{(4x^3)^2}{2x^4}$ (b) $\dfrac{(2b^2)^4}{4b^3}$

4. Simplify. Each expression has two variable bases.
(a) $(a^2b)^3 \times a^4b^2$ (b) $(xy^2)^2 \times x^3y$

5. Simplify.
(a) $\dfrac{(3x^2y)^2}{x^3y}$ (b) $\dfrac{(5a^3b^2)^2}{5a^4b^3}$

Part B — Two Brackets and Three-Step Simplification

6. Simplify. There are two brackets to clear before you combine bases.
(a) $(2x^3)^2 \times (3x)^2$ (b) $(a^2b)^2 \times (ab^2)^3$

7. Simplify. These combine all three steps of the checklist.
(a) $\dfrac{(2x^2)^3 \times x^4}{4x^5}$ (b) $\dfrac{(3a^2b)^2 \times ab^3}{9a^3b^2}$

8. Simplify. Simplify inside the brackets first, then apply the outer power.
(a) $\left(\dfrac{8x^5}{2x^2}\right)^2$ (b) $\left(\dfrac{12a^6b^3}{4a^2b}\right)^2$

9. Simplify. Simplify inside the brackets, apply the outer power, then multiply once more.
(a) $\left(\dfrac{9x^4y^2}{3xy}\right)^2 \times x^2$ (b) $\left(\dfrac{4a^3b}{2ab}\right)^3 \times b^2$

Part C — Mixed Practice

10. Simplify $(3x^2y^3)^2 \times (2xy)^3$.

11. Simplify $\dfrac{(5a^4b^2)^2}{25a^3b^3} \times a^2b$.

12. *(Multiple Choice)* The simplified form of $(2x^2)^3 \times x$ is
A. $2x^7$ B. $8x^6$ C. $8x^7$ D. $6x^7$

13. *(Evaluate)* Simplify $\left(\dfrac{2^3 \times 2^4}{2^5}\right)^2$, then evaluate the result as a single number.

14. *(Challenge)* Simplify $\dfrac{(3a^2b)^2(2ab^3)}{6a^3b^2}$.

Answer Key. 1. (a) x^7 (b) y^5 (c) a^8 (d) $8b^3$. **2.** (a) $4x^7$ (b) $18a^8$. **3.** (a) $8x^2$ (b) $4b^5$. **4.** (a) $a^{10}b^5$ (b) x^5y^5. **5.** (a) $9xy$ (b) $5a^2b$. **6.** (a) $36x^8$ (b) a^7b^8. **7.** (a) $2x^5$ (b) a^2b^3. **8.** (a) $16x^6$ (b) $9a^8b^4$. **9.** (a) $9x^8y^2$ (b) $8a^6b^2$. **10.** $72x^7y^9$. **11.** a^7b^2. **12.** C. **13.** 16. **14.** $3a^2b^3$.

Lesson 5 Assignment *Show your work.*

Part A — Zero Exponents with Numbers

1. Evaluate each power without a calculator.
(a) 6^0 (b) 12^0 (c) $(-9)^0$ (d) $(-12)^0$ (e) 1^0

2. Use the Quotient Law to show that $8^5 \div 8^5 = 1$, and confirm this matches the value of 8^0.

3. Evaluate without a calculator, applying the Zero Exponent Law where needed.
(a) $3^2 \times 3^0$ (b) $5^0 \times 5^3$ (c) $10^0 \times 10^0$ (d) $\dfrac{7^3}{7^3}$ (e) $(2^4)^0$

4. State whether each statement is true or false.
(a) $y^0 = y$ (b) $5x^0 = 5$ (c) $(5x)^0 = 5$ (d) $-2^0 = -1$ (e) $(-2)^0 = 1$

5. Evaluate without a calculator, watching carefully where the exponent applies.
(a) -7^0 (b) $(-7)^0$ (c) -3×4^0 (d) $(-3 \times 4)^0$ (e) $[-(5)^0]^2$

Part B — Variables and Combining the Laws

6. Simplify each expression.
(a) p^0 (b) $8q^0$ (c) $(8q)^0$ (d) $\dfrac{3}{r^0}$ (e) $(4s^2)^0$

7. Simplify each expression.
(a) $x^4 \times x^0$ (b) $\dfrac{y^5}{y^5}$ (c) $6a^0 \times 3a^2$ (d) $\dfrac{12b^3}{b^3}$

8. Simplify each expression.
(a) $(3x^2)^0 \times 5x^3$ (b) $2c^4 \times (5c^2)^0$ (c) $\dfrac{9d^6}{(4d^3)^0 \times d^6}$ (d) $(7e^0)^2$

9. Simplify, combining the Zero Exponent Law with the laws from Lessons 1–4.
(a) $(3x^2y^0)^2 \times 2x^3$ (b) $\dfrac{(2a^4b^0)^3}{4a^5}$

10. Simplify $\dfrac{(4x^2y^0)^2 \times 3x}{6x^3}$.

Part C — Multiple Choice and Explanation

11. *(Multiple Choice)* The value of $\dfrac{6^0 + 3^0}{2^0}$ is
A. 0 B. 1 C. 2 D. 6

12. *(Multiple Choice)* Which expression is equal to 5, assuming $x \neq 0$?
A. $5x^0$ B. $(5x)^0$ C. $0x^5$ D. $5^0 x$

13. *(Explain)* Explain the difference between $9y^0$ and $(9y)^0$, assuming $y \neq 0$, and state the value of each.

Answer Key. 1. (a) 1 (b) 1 (c) 1 (d) 1 (e) 1. **2.** $8^5 \div 8^5 = 8^{5-5} = 8^0$, and any nonzero number divided by itself is 1, so $8^0 = 1$. **3.** (a) 9 (b) 125 (c) 1 (d) 1 (e) 1. **4.** (a) F (b) T (c) F (d) T (e) T. **5.** (a) -1 (b) 1 (c) -3 (d) 1 (e) 1. **6.** (a) 1 (b) 8 (c) 1 (d) 3 (e) 1. **7.** (a) x^4 (b) 1 (c) $18a^2$ (d) 12. **8.** (a) $5x^3$ (b) $2c^4$ (c) 9 (d) 49. **9.** (a) $18x^7$ (b) $2a^7$. **10.** $8x^2$. **11.** C. **12.** A. **13.** $9y^0 = 9 \times 1 = 9$ (exponent on y only); $(9y)^0 = 1$ (exponent on the whole product).

Lesson 6 Assignment *Show your work.*

Part A — The Negative Exponent Rule

1. Write each power with a positive exponent. (Do not evaluate.)
(a) x^{-5} (b) y^{-9} (c) a^{-1} (d) c^{-k}

2. Write each power with a positive exponent, then evaluate without a calculator.
(a) 2^{-4} (b) 6^{-1} (c) 3^{-3} (d) 10^{-3}

3. Simplify each expression so the exponent is positive.
(a) $\dfrac{1}{4^{-2}}$ (b) $\dfrac{1}{9^{-1}}$ (c) $\dfrac{1}{x^{-5}}$ (d) $\dfrac{1}{y^{-8}}$

4. Evaluate without a calculator.
(a) 5^{-2} (b) 2^{-5} (c) 7^{-1} (d) 4^{-3}

5. State whether each statement is True or False.
(a) $3x^{-2} = \frac{1}{3x^2}$ (b) $5y^{-4} = \frac{5}{y^4}$ (c) $\frac{1}{a^{-3}} = a^3$ (d) $\frac{2}{b^{-1}} = 2b$ (e) $x^{-1} = -x$

Part B — Coefficients, the Quotient Law and Applications

6. Simplify each expression, writing the answer with a positive exponent.
(a) $4x^{-3}$ (b) $9a^{-1}$ (c) $\dfrac{6}{y^{-2}}$ (d) $\dfrac{12}{b^{-5}}$

7. Evaluate without a calculator. Be careful where the negative sign is attached.
(a) $(-3)^{-2}$ (b) -3^{-2} (c) $(-2)^{-3}$ (d) -2^{-3}

8. Simplify using the quotient law, then write the answer with a positive exponent.
(a) $x^2 \div x^6$ (b) $a^3 \div a^7$ (c) $m \div m^5$ (d) $y^4 \div y^9$

9. Simplify using the quotient law, then write the answer with a positive exponent.
(a) $\dfrac{10x^2}{5x^7}$ (b) $\dfrac{18a^3}{6a^8}$ (c) $\dfrac{8y}{2y^6}$ (d) $\dfrac{20b^2}{4b^9}$

10. Simplify each expression, writing the answer with a positive exponent.
(a) $\dfrac{15a^2b^4}{3a^6b}$ (b) $\dfrac{24x^3y}{6x^5y^4}$

11. A student simplified 6^{-2} and wrote $6^{-2} = -12$. Explain what mistake the student made, and state the correct value of 6^{-2}.

12. *(Multiple Choice)* The value of $2^{-3} + 2^0$ is A. -1 B. 0 C. 9/8 D. 1/8

13. *(Multiple Choice)* Which of the following is equivalent to $\dfrac{7}{x^{-4}}$? A. $7x^4$ B. $\dfrac{7}{x^4}$ C. $\dfrac{x^4}{7}$ D. $-7x^4$

Answer Key. 1. (a) $1/x^5$ (b) $1/y^9$ (c) $1/a$ (d) $1/c^k$. **2.** (a) 1/16 (b) 1/6 (c) 1/27 (d) 1/1000. **3.** (a) 16 (b) 9 (c) x^5 (d) y^8. **4.** (a) 1/25 (b) 1/32 (c) 1/7 (d) 1/64. **5.** (a) F (b) T (c) T (d) T (e) F. **6.** (a) $4/x^3$ (b) $9/a$ (c) $6y^2$ (d) $12b^5$. **7.** (a) 1/9 (b) $-1/9$ (c) $-1/8$ (d) $-1/8$. **8.** (a) $1/x^4$ (b) $1/a^4$ (c) $1/m^4$ (d) $1/y^5$. **9.** (a) $2/x^5$ (b) $3/a^5$ (c) $4/y^5$ (d) $5/b^7$. **10.** (a) $5b^3/a^4$ (b) $4/(x^2y^3)$. **11.** The negative exponent means *reciprocal*, not "multiply base by exponent" and not a negative answer; $6^{-2} = 1/6^2 = 1/36$. **12.** C. **13.** A.

Lesson 7 Assignment *Show your work.*

Part A — Warm-Up: One Law at a Time

1. Simplify each expression using just *one* law. Write with positive exponents.
(a) $x^5 \times x^{-8}$ (b) $y^{-4} \times y^{-3}$ (c) $(a^{-3})^2$ (d) $b^0 \times b^6$

2. Simplify using the quotient law. Write with positive exponents.
(a) $x^3 \div x^9$ (b) $18a^{-2} \div 6a^3$

3. Evaluate. Watch carefully what the exponent 0 is actually attached to.
(a) $(7x^4)^0$ (b) $7x^0$

4. Rewrite each expression with positive exponents only.
(a) x^{-6} (b) $5a^{-2}$ (c) $(3y)^{-2}$ (d) $\frac{1}{c^{-4}}$

Part B — Two or More Laws Together

5. Simplify. Apply the power law first, then the product law.
(a) $(2x^2)^3 \times x^{-4}$ (b) $(3a^{-2})^2 \times a^5$

6. Simplify. Apply the power law first, then the quotient law.
(a) $\frac{(2x^3)^2}{x^7}$ (b) $\frac{(4y^{-1})^2}{2y^3}$

7. Simplify. Each expression has two variable bases.
(a) $\frac{a^4 b^{-2}}{a^{-3} b^5}$ (b) $x^{-2}y^3 \times x^5 y^{-6}$

8. Simplify inside the brackets first, then apply the outer power.
(a) $\left(\frac{x^5}{x^2}\right)^{-2}$ (b) $\left(\frac{6a^4}{2a}\right)^{-1}$

9. Simplify. These combine all three steps of the checklist.
(a) $\frac{(2x^2)^3 \times x^{-1}}{4x^4}$ (b) $\frac{(3a^3 b)^2}{9a^4 b^{-1}}$

10. Simplify or evaluate without a calculator.
(a) $\left(\frac{2}{3}\right)^{-2}$ (b) $\left(\frac{x}{4}\right)^{-1}$

11. Simplify inside the brackets first, then apply the outer power.

(a) $\left(\frac{a^3}{2a^{-1}}\right)^{-2}$

(b) $\frac{(x^{-2}y)^3}{x^{-4}y^2}$

12. Simplify $\dfrac{(3x^2y^0)^{-1} \times 6x^5}{2x^{-3}}$.

Part C — Mixed Practice Test

13. Simplify $\frac{(2a^3b^{-2})^2}{4a^{-1}b^3}$.

14. Simplify $(x^{-3}y^2)^{-2} \times \frac{x^4}{y^5}$.

15. *(Multiple Choice)* The simplified form of $\frac{(2x^2)^{-1}}{x^3}$ is
A. $2x^5$ B. $\frac{1}{2x^5}$ C. $\frac{1}{2x^{-5}}$ D. $-2x^5$

16. *(Evaluate)* Evaluate $\frac{5^{-2}\times 5^4}{5^0}$ without using a calculator.

Answer Key. 1. (a) $1/x^3$ (b) $1/y^7$ (c) $1/a^6$ (d) b^6. **2.** (a) $1/x^6$ (b) $3/a^5$. **3.** (a) 1 (b) 7. **4.** (a) $1/x^6$ (b) $5/a^2$ (c) $1/(9y^2)$ (d) c^4. **5.** (a) $8x^2$ (b) $9a$. **6.** (a) $4/x$ (b) $8/y^5$. **7.** (a) a^7/b^7 (b) x^3/y^3. **8.** (a) $1/x^6$ (b) $1/(3a^3)$. **9.** (a) $2x$ (b) a^2b^3. **10.** (a) 9/4 (b) $4/x$. **11.** (a) $4/a^8$ (b) y/x^2. **12.** x^6. **13.** a^7/b^7. **14.** x^{10}/y^9. **15.** B. **16.** 25.

Lesson 8 Assignment *Show your work.*

Part A — Converting Between Forms

1. Complete the table below, converting each number to the missing form.

Standard Notation	Scientific Notation
91 000	
0.00027	
5.9	
	6.3×10^5
	4×10^{-3}
	8.15×10^2

2. Write each number in scientific notation.
(a) 8 000 (b) 43 000 (c) 610 000 (d) 7 250 000 (e) 96

3. Write each number in scientific notation.
(a) 0.005 (b) 0.00061 (c) 0.0000034 (d) 0.08 (e) 0.000009

4. Express each quantity in scientific notation.
(a) At its closest, Venus comes within about 41 000 000 km of Earth. (b) A single grain of table salt has a mass of about 0.000065 g. (c) A country's annual budget was about 5 600 000 000 dollars.

5. Write each number in standard notation.
(a) 4×10^3 (b) 6.7×10^4 (c) 9.02×10^6 (d) 3×10^8

6. Write each number in standard notation.
(a) 5×10^{-2} (b) 2.3×10^{-4} (c) 8×10^{-6} (d) 1.9×10^{-3}

7. Express each quantity in standard notation.
(a) The diameter of a certain bacterium is about 4×10^{-6} m. (b) A city's population is about 2.15×10^6 people.

Part B — Reasoning, Comparing and Ordering

8. Determine the direction, and how many places the decimal point should be moved, to write each number in scientific notation.
(a) 52 (b) 306 000 (c) 0.00041 (d) 0.7

9. Determine whether each number below is correctly written in scientific notation. If it is not, explain why and rewrite it correctly.

(a) 6.2×10^4

(b) 47×10^3

(c) 0.8×10^{-2}

(d) 1×10^0

10. Order each set of numbers from smallest to largest.
(a) $6 \times 10^{-5}, \quad 3 \times 10^{-3}, \quad 9 \times 10^{-7}$

(b) $2.1 \times 10^4, \quad 5 \times 10^3, \quad 8.6 \times 10^4$

11. Convert each number to the other form (scientific notation $\leftrightarrow$ standard notation).
(a) 27 000

(b) 0.00033

(c) 4.8×10^5

(d) 9×10^{-4}

(e) 500 000 000

(f) 0.0000012

12. A virus is measured at 5×10^{-7} m across, while a bacterium is measured at 3×10^{-6} m across. Which is larger? Justify your answer by comparing the exponents.

13. *(Multiple Choice)* The distance across a certain crystal lattice is approximately 0.00000038 m. Written in scientific notation as $a \times 10^n$, the value of n is A. 7 B. -7 C. 6 D. -6

14. *(Multiple Choice)* In 2024, a company's total sales were approximately 750 million dollars. In scientific notation, the number of dollars is A. 7.5×10^2 B. 7.5×10^5 C. 7.5×10^8 D. 7.5×10^{11}

15. *(Numerical Response)* A number written in scientific notation is $a \times 10^n$, where $a = 6.4$. If the number in standard form is 640 000, the value of n is _____.

Answer Key. 1. 9.1×10^4; 2.7×10^{-4}; 5.9×10^0; 630 000; 0.004; 815. **2.** (a) 8×10^3 (b) 4.3×10^4 (c) 6.1×10^5 (d) 7.25×10^6 (e) 9.6×10^1. **3.** (a) 5×10^{-3} (b) 6.1×10^{-4} (c) 3.4×10^{-6} (d) 8×10^{-2} (e) 9×10^{-6}. **4.** (a) 4.1×10^7 km (b) 6.5×10^{-5} g (c) 5.6×10^9 dollars. **5.** (a) 4 000 (b) 67 000 (c) 9 020 000 (d) 300 000 000. **6.** (a) 0.05 (b) 0.00023 (c) 0.000008 (d) 0.0019. **7.** (a) 0.000004 m (b) 2 150 000 people. **8.** (a) 1 left (b) 5 left (c) 4 right (d) 1 right. **9.** (a) correct ($1 \leq 6.2 < 10$) (b) incorrect, $47 \geq 10$; 4.7×10^4 (c) incorrect, $0.8 < 1$; 8×10^{-3} (d) correct. **10.** (a) $9 \times 10^{-7} < 6 \times 10^{-5} < 3 \times 10^{-3}$ (b) $5 \times 10^3 < 2.1 \times 10^4 < 8.6 \times 10^4$. **11.** (a) 2.7×10^4 (b) 3.3×10^{-4} (c) 480 000 (d) 0.0009 (e) 5×10^8 (f) 1.2×10^{-6}. **12.** Bacterium, since $-6 > -7$. **13.** B. **14.** C. **15.** 5.

Lesson 9 Assignment *Show your work.*

Part A — Evaluating Rational-Exponent Powers

1. Evaluate without using a calculator.
(a) $64^{1/2}$ (b) $121^{1/2}$ (c) $125^{1/3}$ (d) $1\,000^{1/3}$ (e) $81^{1/4}$ (f) $32^{1/5}$

2. Evaluate without using a calculator. Take the root first to keep the numbers small.
(a) $8^{4/3}$ (b) $9^{5/2}$ (c) $25^{3/2}$ (d) $16^{5/4}$ (e) $64^{2/3}$ (f) $4^{7/2}$

3. Evaluate without using a calculator.
(a) $\left(\frac{4}{9}\right)^{1/2}$ (b) $\left(\frac{8}{27}\right)^{1/3}$ (c) $\left(\frac{16}{81}\right)^{3/4}$

Part B — Converting Between Radicals and Rational Exponents

4. Write an equivalent expression using radicals.
(a) $p^{1/6}$ (b) $q^{1/7}$ (c) $r^{2/5}$ (d) $s^{3/8}$ (e) $t^{-1/3}$ (f) $u^{-3/4}$

5. Write each radical as a power in the form k^n, where $n \in \mathbb{Q}$.
(a) $\sqrt[4]{p^3}$ (b) $\sqrt[5]{q^2}$ (c) $\sqrt{r^9}$ (d) $\sqrt[3]{s}$ (e) $\frac{1}{\sqrt[5]{t^5}}$ (f) $\frac{1}{\sqrt{u^3}}$

Part C — Exponent Laws with Rational Exponents

6. Simplify. Leave your answer as a power with a positive exponent. (Product Law.)
(a) $m^{1/3} \times m^{1/6}$ (b) $n^{2/5} \times n^{1/5}$ (c) $p^{1/4} \times p^{3/4}$ (d) $q^{1/2} \times q^{1/2}$

7. Simplify. Leave your answer as a power with a positive exponent. (Quotient Law.)
(a) $m^{7/10} \div m^{1/10}$ (b) $n^{3/4} \div n^{1/4}$ (c) $p^{7/8} \div p^{3/8}$ (d) $q^{1/2} \div q^{3/2}$

8. Simplify. Leave your answer as a power with a positive exponent. (Power of a Power, Power of a Product.)
(a) $\left(m^{1/2}\right)^4$ (b) $\left(n^{1/3}\right)^9$ (c) $\left(3p^{1/2}\right)^2$ (d) $\left(9q^{2/3}\right)^{3/2}$

Part D — Applications and Extensions

9. A cube has a volume of 27 cm^3.
(a) Write a power that represents the edge length of the cube. (b) Write a power that represents the surface area of the cube. (c) Calculate the exact edge length and surface area of the cube.

10. A cube has a volume of W cm^3.
(a) Write a power *and* a radical that represent the edge length of the cube.
(b) Write a power *and* a radical that represent the area of one face of the cube.

11. In each case, write the given number as a power with the given exponent.
(a) 6, exponent $\frac{1}{2}$ (b) 3, exponent $\frac{1}{4}$ (c) 5, exponent $\frac{1}{3}$ (d) $\frac{1}{4}$, exponent $-\frac{1}{2}$

12. *(Multiple Choice)* $\left(\frac{9}{4}\right)^{-1/2}$ is equal to A. $\frac{3}{2}$ B. $\frac{2}{3}$ C. $\frac{4}{9}$ D. $-\frac{2}{3}$

13. *(Multiple Choice)* Which expression is *not* equivalent to $x^{3/4}$? A. $\sqrt[4]{x^3}$ B. $\left(\sqrt[4]{x}\right)^3$ C. $\left(\sqrt{x}\right)^{3/2}$ D. $\sqrt[3]{x^4}$

14. *(Numerical Response)* Find the exact value of $\left(x^{1/2}+y^{2/3}\right)^2$ when $x=16$ and $y=8$.

15. Simplify $a^{2/3}\times a^{1/4}$. Leave your answer as a power with a positive exponent.

16. Simplify $2b^{1/2}\times 3b^{1/4}$. Write your answer as a power with a positive exponent, and then in radical form.

Answer Key. 1. (a) 8 (b) 11 (c) 5 (d) 10 (e) 3 (f) 2. **2.** (a) 16 (b) 243 (c) 125 (d) 32 (e) 16 (f) 128. **3.** (a) 2/3 (b) 2/3 (c) 8/27. **4.** (a) $\sqrt[6]{p}$ (b) $\sqrt[7]{q}$ (c) $\sqrt[5]{r^2}$ (d) $\sqrt[8]{s^3}$ (e) $1/\sqrt[3]{t}$ (f) $1/\sqrt[4]{u^3}$. **5.** (a) $p^{3/4}$ (b) $q^{2/5}$ (c) $r^{9/2}$ (d) $s^{1/3}$ (e) $t^{-5/6}$ (f) $u^{-3/2}$. **6.** (a) $m^{1/2}$ (b) $n^{3/5}$ (c) p (d) q. **7.** (a) $m^{3/5}$ (b) $n^{1/2}$ (c) $p^{1/2}$ (d) $1/q$. **8.** (a) m^2 (b) n^3 (c) $9p$ (d) $27q$. **9.** (a) $27^{1/3}$ (b) $6\left(27^{1/3}\right)^2$ (c) edge 3 cm, surface area 54 cm^2. **10.** (a) $W^{1/3}=\sqrt[3]{W}$ (b) $W^{2/3}=\left(\sqrt[3]{W}\right)^2$. **11.** (a) $36^{1/2}$ (b) $81^{1/4}$ (c) $125^{1/3}$ (d) $16^{-1/2}$. **12.** B. **13.** D. **14.** 64. **15.** $a^{11/12}$. **16.** $6b^{3/4}=6\sqrt[4]{b^3}$.

Lesson 10 Assignment *Show your work.*

Part A — The Exponent Laws

1. Evaluate using the product law.
(a) $5^2 \times 5^2$ (b) $2^3 \times 2^4$

2. Simplify.
(a) $8t^3 \times t^5$ (b) $k^4 \times k \times k^2$

3. Simplify.
(a) $4g^2 \times 5g^3$ (b) $3h \times 7h^4$

4. Simplify using the quotient law.
(a) $\dfrac{9^6}{9^4}$ (b) $s^{12} \div s^5$

5. Simplify using the power of a power law.
(a) $(3^2)^3$ (b) $(w^4)^3$

6. Simplify $\dfrac{(s^3)^4}{s^5}$, using both the power of a power law and the quotient law.

7. Simplify.
(a) $(5g)^2$ (b) $(2r)^4$

8. Simplify.
(a) $\left(\dfrac{h}{3}\right)^2$ (b) $\left(\dfrac{3}{k}\right)^3$

9. Simplify.
(a) $(-3t)^2$ (b) $(2ef)^3$

Part B — Combining Laws, Zero and Negative Exponents

10. Simplify $(3t^2)^2 \times t^4$.

11. Simplify $\dfrac{(2w^3)^3}{4w^5}$.

12. Evaluate.
(a) 7^0 (b) $(-4)^0$

13. Simplify.

(a) $9g^0$

(b) $(9g)^0$

14. Evaluate.

(a) 6^{-2}

(b) $\dfrac{1}{4^{-3}}$

15. Simplify, writing each answer with a positive exponent.

(a) $8t^{-2}$

(b) $\dfrac{5}{k^{-4}}$

16. Simplify $\dfrac{20w^{-3}}{5w^2}$. Write the answer with a positive exponent.

17. Simplify $(2t^{-2})^3 \times t^4$. Write the answer with a positive exponent.

Part C — Scientific Notation and Rational Exponents

18. Write each number in scientific notation.

(a) 58 000

(b) 0.00047

19. Write each number in standard notation.

(a) 6.3×10^4

(b) 9×10^{-5}

20. A dust mite is about 2.0×10^{-4} m long. About how many dust mites, laid end to end, would span a length of 0.6 m?

21. Express each power in radical form.

(a) $t^{2/5}$

(b) $g^{5/6}$

22. *(Numerical Response)* If $p = 36$ and $q = -125$, the value of $p^{3/2} + q^{2/3}$ is _____.

Part D — Mixed Challenge

23. Simplify $\dfrac{(3t^2)^{-1} \times 6t^5}{2t^{-1}}$, expressing your answer with a positive exponent.

24. Simplify $s^{7/4} \times s^{-3/4}$. Then, if $s = 810\,000$, express the numeric value of the result in scientific notation.

25. *(Written Response)* A square data-storage chip has an area of $(5 \times 10^2)^2$ mm^2.

(a) Express the area of the chip in proper scientific notation.

(b) The chip contains 5.0×10^7 transistors, spread evenly over its surface. Determine how many transistors there are per square millimetre of area.

Answer Key. 1. (a) 625 (b) 128. **2.** (a) $8t^8$ (b) k^7. **3.** (a) $20g^5$ (b) $21h^5$. **4.** (a) 81 (b) s^7. **5.** (a) 729 (b) w^{12}. **6.** s^7. **7.** (a) $25g^2$ (b) $16r^4$. **8.** (a) $h^2/9$ (b) $27/k^3$. **9.** (a) $9t^2$ (b) $8e^3f^3$. **10.** $9t^8$. **11.** $2w^4$. **12.** (a) 1 (b) 1. **13.** (a) 9 (b) 1. **14.** (a) 1/36 (b) 64. **15.** (a) $8/t^2$ (b) $5k^4$. **16.** $4/w^5$. **17.** $8/t^2$. **18.** (a) 5.8×10^4 (b) 4.7×10^{-4}. **19.** (a) 63 000 (b) 0.00009. **20.** 3 000 mites. **21.** (a) $\sqrt[5]{t^2}$ (b) $\sqrt[6]{g^5}$. **22.** 241. **23.** t^4. **24.** s; 8.1×10^5. **25.** 2.5×10^5 mm^2; 200 transistors per mm^2.